

WASP-52b

R-0443

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_171102 · created 2026-07-19T14:46:34+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458059.6437 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.032
Transit depth (Rp/Rs)^2	1.78 +/- 0.85 [%]
Orbital Inclination (inc)	84.69 +/- 1.91
Airmass coefficient 1 (a1)	1.067 +/- 0.018
Airmass coefficient 2 (a2)	-0.055 +/- 0.052
Scatter in the residuals of the lightcurve fit is	1.66 %
Best Comparison Star	None
Optimal Aperture	3.99
Optimal Annulus	12.65
Transit Duration (day)	0.063 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.133 ± 0.032 vs 0.1646 (deviation 0.99σ)
Duration fit vs published	0.063 d vs 0.0758 d (deviation 0.64σ)
Mid-time O–C	-6.7 min
Depth SNR	4.2
Residual scatter	1.66 %

Light curve

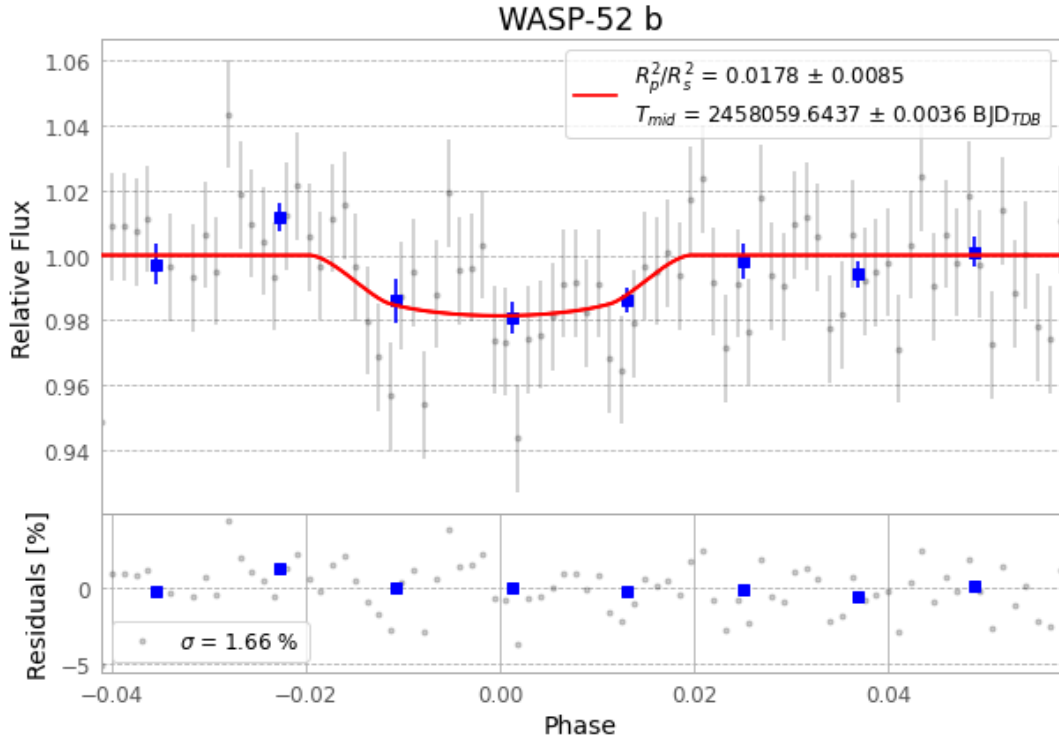


Figure 1 — FinalLightCurve_WASP-52 b_2017-11-01.png (SHA-256 5b489c959f88c672...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 217], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d493f38099e849ab...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

84 FITS frames (55 MB), WASP-52171102013612.FITS → WASP-52171102054512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-171102.log (1a4f4693f07f...), FinalLightCurve_WASP-52 b_2017-11-01.png (5b489c959f88...), FinalLightCurve_WASP-52 b_2017-11-01.csv (3f38901352c6...)

Compute resources

Wall time 160.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 14:46Z.